

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: **Deep Learning**

COURSE CODE: **MLA04**

COURSE OUTCOME COVERED

C05: Compare and assess emerging technologies including AI/ML and Blockchain with respect to their applicability, societal impact, ethical considerations, and sustainability in the context of modern software system design and development. (BL5)

Assessment Tool Weightage:

QUESTIONS:5

Total Marks:25

ANNEXURE A

ERROR ANALYSIS TASK

Code of Conduct:

*I __Sk. Fawaz Ahamad/192425390__ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

ANNEXURE A

A CNN model was trained to classify images into four categories: **Cats, Dogs, Birds, and Rabbits**. The model was tested on **200 images**, producing the following confusion matrix.

Actual \ Predicted	Cat	Dog	Bird	Rabbit
Cat	42	5	2	1
Dog	6	38	3	3
Bird	1	4	43	2
Rabbit	2	3	5	40

Answer the following questions:

1. Calculate the **overall accuracy** of the model.
2. Identify **which class has the highest number of misclassifications**.
3. List the **two most common classification errors** made by the model.
4. Suggest **three possible reasons** why these errors occurred.
5. Recommend **four techniques** to improve the model performance.
6. Explain whether **collecting more training data** or **increasing the number of CNN layers** would be more beneficial in this case. Justify your answer.
7. If the model achieves **99% training accuracy** but only **82% testing accuracy**, what problem does the model suffer from? Explain how you would address it.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Error Identification	20%	Accurately identifies all errors with clear understanding of issues	Identifies most errors with minor omissions	Identifies limited errors; some are missed	Fails to identify key errors
Root Cause Analysis	30%	Clearly explains underlying causes with strong technical reasoning	Explains causes with moderate clarity	Partial explanation; weak reasoning	Unable to identify causes of errors
Correction & Justification	25%	Provides correct solutions with strong justification and logic	Provides correct corrections with some justification	Corrections are partially correct or weakly justified	Incorrect or no corrections provided
Application of Concepts	15%	Effectively applies relevant concepts to analyze and resolve errors	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts
Clarity & Presentation	10%	Presents analysis clearly, logically, and systematically	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand

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Course name: - Deep Learning

ANNEXURE A

Q. A CNN model was trained to classify images into four categories: Cats, Dogs, Birds, and Rabbits. The model was tested on 200 images, producing the following confusion matrix.

Q1. Calculate the Overall Accuracy?

Formula: -

$$\text{Accuracy} = (\text{Correct Predictions} / \text{Total Predictions}) \times 100$$

Correct predictions are the diagonal values:

- Cat = 42
- Dog = 38
- Bird = 43
- Rabbit = 40

Total Correct

$$= 42 + 38 + 43 + 40$$

$$= 163$$

Total Images = **200**

Accuracy

$$= (163 / 200) \times 100$$

$$= 81.5\%$$

Answer: 81.5%

Q2. Which class has the highest number of misclassifications?

Misclassifications = Row Total – Correct Prediction

Cat

$$50 - 42 = 8$$

Dog

$$50 - 38 = 12$$

Bird

$$50 - 43 = 7$$

Rabbit

$$50 - 40 = 10$$

Class	Misclassifications
Cat	8
Dog	12
Bird	7
Rabbit	10

Answer: Dog has the highest number of misclassifications (12).

Q3. Two most common classification errors?

Look for the largest off-diagonal values.

- Dog predicted as Cat = 6
- Rabbit predicted as Bird = 5
- Cat predicted as Dog = 5

Since the question asks for **two**, you can write:

- Dog $\rightarrow$ Cat (**6**)
- Rabbit $\rightarrow$ Bird (**5**)

(We may also mention Cat $\rightarrow$ Dog = 5.)

Q4. Three possible reasons for these errors?

1. Similar visual features between classes (e.g., cats and dogs).
2. Insufficient or imbalanced training data.
3. Poor image quality, lighting, or background noise.

Q5. Four techniques to improve model performance?

1. Collect more diverse training images.
2. Apply data augmentation (rotation, flipping, zooming).
3. Tune CNN hyperparameters (learning rate, batch size, epochs).
4. Use transfer learning with pretrained models such as Resnet or VGG.

Q6. More training data or more CNN layers?

Answer: More training data.

Justification

- The current errors suggest the model needs better examples of each class.
- More data helps the model generalize better.
- Simply adding more CNN layers increases complexity and may lead to overfitting if the dataset is limited.

Q7. Training accuracy = 99%, Testing accuracy = 82%?

Problem

The model is suffering from **Overfitting**.

Reason

- It memorizes the training data.
- It does not generalize well to unseen test data.

How to solve it

- Use data augmentation.
- Add dropout layers.
- Apply L2 regularization.
- Use early stopping.
- Collect more training data.
- Reduce model complexity if necessary.